PRE-UNIVERSITY 1
COMPUTER SCIENCE
Standard and Higher Level

27 March 2026

1 hour

Instructions to Candidates

- Do not open this examination paper until instructed to do so.
- Answer ALL questions in the provided box.
- The maximum mark for this examination paper is **40 marks.**

Name: _____ Date: _____ Marks: _____/ 40

Summative Assessment 2.3

1. a) Outline two differences between **Random Access Memory (RAM) and Read-Only Memory (ROM).** [4]
   b) Outline the function of an accumulator. [2]

2. **A computer system uses various components to process data and instructions.**
   a) Outline the function of the Control Unit (CU) in a CPU. [2]
   b) State two differences between volatile and non-volatile memory. [2]
   c) Explain why cache memory improves system performance. [2]

3. **Describe the role of the following buses in the operation of a CPU:**
   a) Control Bus. [2]
   b) Address Bus. [2]

4. **A computer system uses three levels of cache memory (L1, L2, and L3).** Explain the relationship between the location of cache (proximity to the CPU), its storage capacity, and its access speed. [6]

5. **A modern smartphone uses a multi-core processor to handle various applications.**
   a) Outline why a quad-core processor may not always perform a task four times faster than a single-core processor. [3 marks]
   b) Explain the purpose of the **Program Counter (PC)** and the **Memory Address Register (MAR)** during the *Fetch* stage of the instruction cycle. [4 marks]

6. **An architect uses a high-performance computer for 3D rendering and video processing.**
   a) Identify the specialized processor best suited for accelerating image and video rendering. [1]
   b) Explain how the processor identified in (a) enhances system performance when compared to a standard CPU for these specific tasks. [4]
   c) Describe the "Fetch-Decode-Execute" cycle as the fundamental process for executing instructions. [6]